

Relationship between two continuous variables

EDUC 641: Week 6

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Goals

- Visualize & describe relationships between two quantitative variables using scatterplots (shape, trend, strength).
- Define and compute covariance, explain its properties (symmetry, relation to variance), and recognize unit dependence.
- Define, compute, and interpret Pearson's correlation as standardized covariance; compare magnitude, direction, and bounds (-1 to $+1$).
- Test the significance of the correlation using the t -test; compute t -statistic, degrees of freedom, and interpret p -values.
- Use R for correlation analysis: `cor()` and `Hmisc::rcorr()` functions
- Report a correlation matrix in APA style
- Handle missing data in correlations via listwise vs. pairwise deletion, noting pros/cons and implications for sample size and comparability.

Covariance and Correlation

Historical Origin

- **Francis Galton (1822–1911)** is widely credited with introducing the statistical concept of *correlation* in the late 19th century.
- In his 1888 paper “*Co-relations and their Measurement, chiefly from Anthropometric Data*”, Galton used the term “**co-relation or correlation of structure**” and attempted to formally define and measure the degree to which two variables vary together.
- Galton noted that the phrase **co-relation** was already used in biology but had not yet been formally defined or quantified.
- The hyphenated **co-relation** appears in his 1888 paper; over time, it evolved into the modern spelling **correlation**.

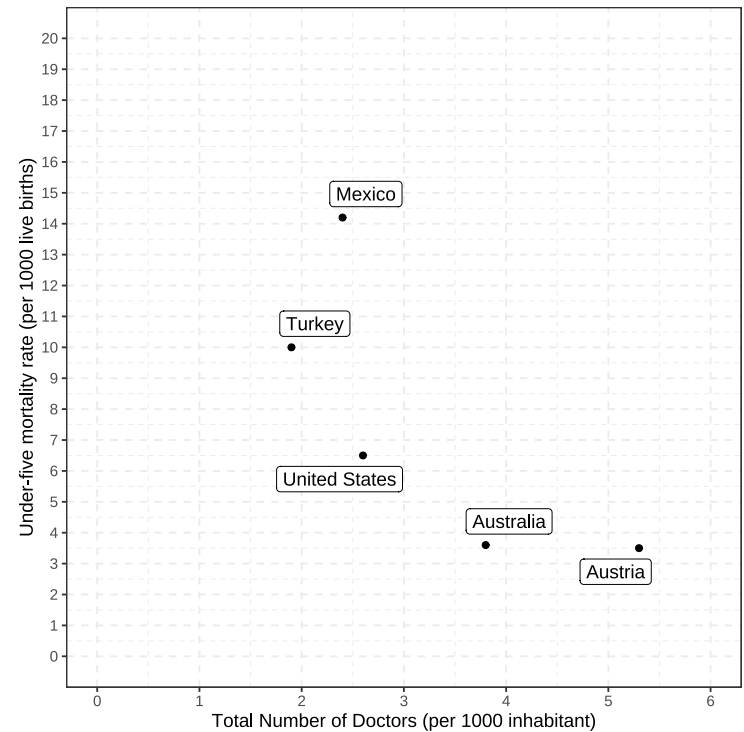
Pearson Correlation

- While Galton introduced and named the idea of co-relation, **Karl Pearson** formalized the **product-moment correlation coefficient** in 1896, building on Galton’s ideas and earlier mathematical work by French physicist Auguste Bravais.
- The **Pearson correlation coefficient** (ρ) is now the most widely used measure of linear association between **continuous variables**.

Scatterplot

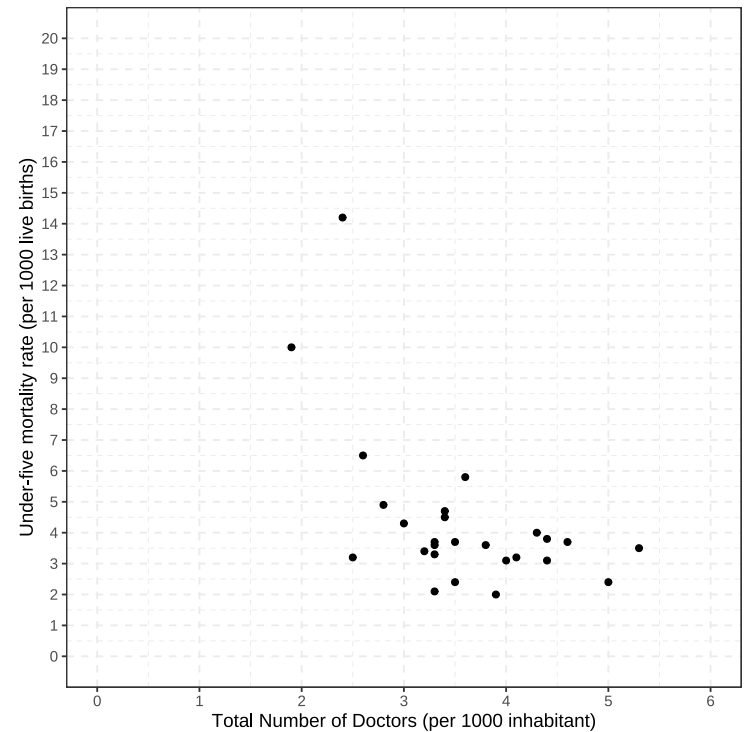
- A **scatterplot** is a graphical tool used to visualize the relationship between two quantitative variables.
- Each point represents a single observation, located according to its values on the two variables, forming (x, y) coordinates.
- By convention, the **predictor** (independent variable) is placed on the **x-axis**, and the **outcome** (dependent or criterion variable) on the **y-axis**.
- Scatterplots help us visually assess:
 - Trend/Direction of the pattern
 - Linearity vs. non-linearity
 - Strength of association
 - Outliers and unusual patterns
 - Possible clusters or heteroscedasticity
- Example variables (2019):
 - Physicians per 1,000 inhabitants
 - Under-five mortality rate per 1,000 live births

country	doctor	infmor
Australia	3.8	3.6
Austria	5.3	3.5
Belgium	3.2	3.4
Canada	2.8	4.9
Czech Republic	4.1	3.2
Estonia	3.5	2.4
France	3.4	4.5
Germany	4.4	3.8
Hungary	3.5	3.7
Iceland	3.9	2
Ireland	3.3	3.3
Israel	3.3	3.7
Italy	4	3.1
Korea	2.5	3.2
Latvia	3.3	3.6
Lithuania	4.6	3.7
Mexico	2.4	14.2
New Zealand	3.4	4.7
Norway	5	2.4
Slovak Republic	3.6	5.8
Slovenia	3.3	2.1
Spain	4.4	3.1
Switzerland	4.3	4
Turkey	1.9	10
United Kingdom	3	4.3
United States	2.6	6.5



Imagine that we represent each country in a two-dimensional space using the pairs of values as X-Y coordinates.

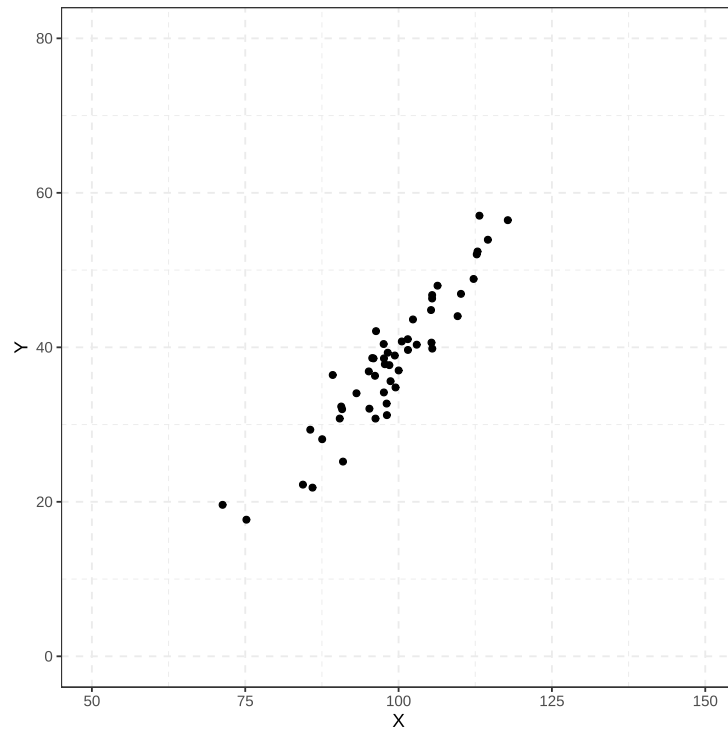
country	doctor	infmor
Australia	3.8	3.6
Austria	5.3	3.5
Belgium	3.2	3.4
Canada	2.8	4.9
Czech Republic	4.1	3.2
Estonia	3.5	2.4
France	3.4	4.5
Germany	4.4	3.8
Hungary	3.5	3.7
Iceland	3.9	2
Ireland	3.3	3.3
Israel	3.3	3.7
Italy	4	3.1
Korea	2.5	3.2
Latvia	3.3	3.6
Lithuania	4.6	3.7
Mexico	2.4	14.2
New Zealand	3.4	4.7
Norway	5	2.4
Slovak Republic	3.6	5.8
Slovenia	3.3	2.1
Spain	4.4	3.1
Switzerland	4.3	4
Turkey	1.9	10
United Kingdom	3	4.3
United States	2.6	6.5



Describing the Pattern in a Scatterplot

When describing the relationship in a scatterplot, consider three features: **shape**, **trend**, and **strength**.

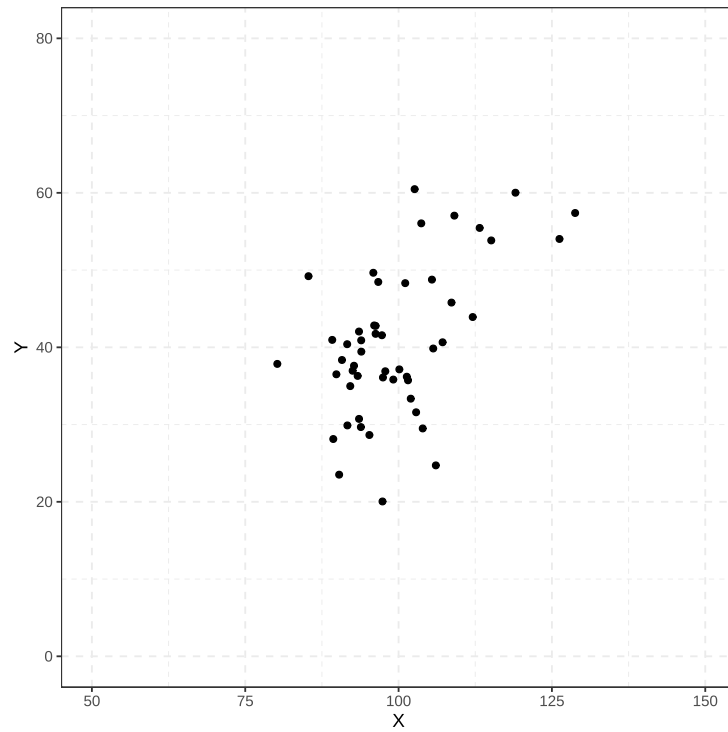
- **Shape:** Describe the form of the relationship:
 - Linear
 - Curvilinear (e.g., quadratic, exponential)
 - Clusters or subgroups
 - Outliers
- **Trend (Direction)**
 - **Positive trend:** Higher values on one variable are associated with higher values on the other.
 - **Negative trend:** Higher values on one variable are associated with lower values on the other.
 - **No trend:** No consistent direction in the relationship.
- **Strength**
 - **Strong:** Points lie close to an imaginary line or curve.
 - **Weak:** Points are widely scattered.
 - Consider whether the strength is **constant or changes** across the range (e.g., heteroscedasticity).



Shape:

Trend:

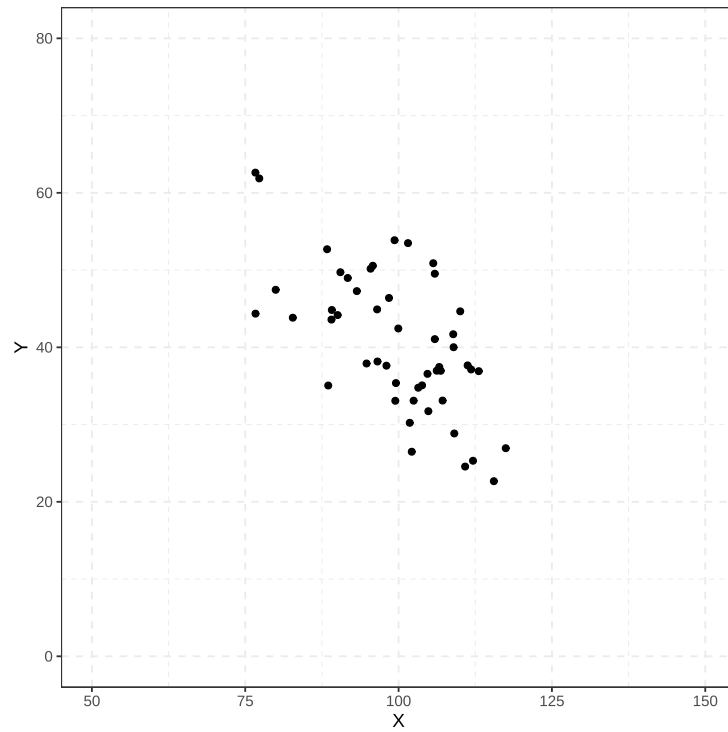
Strength:



Shape:

Trend:

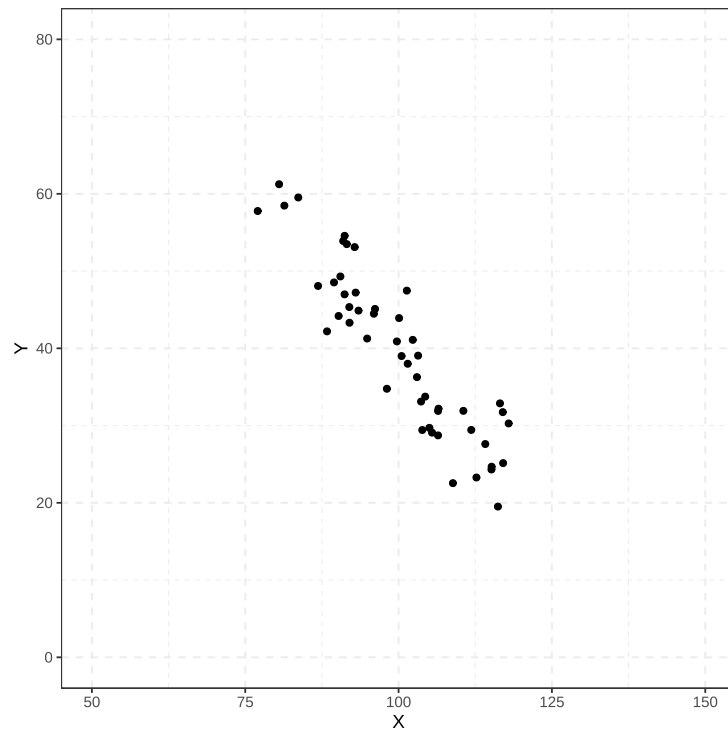
Strength:



Shape:

Trend:

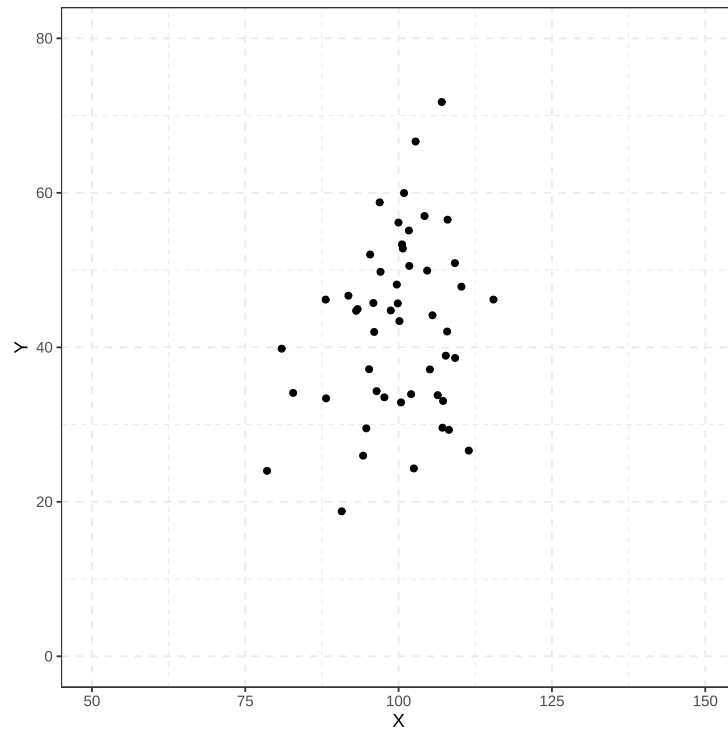
Strength:



Shape:

Trend:

Strength:



Shape:

Trend:

Strength:

Covariance

- **Covariance** is a numerical measure of how two variables co-vary, or how they change together relative to their means.
 - If both variables tend to be above (or below) their means together, the covariance is positive.
 - If one tends to be above its mean when the other is below, the covariance is negative.
- The sample covariance between (X) and (Y) is:

$$Cov(X, Y) = \frac{1}{N-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$$

- This quantity represents the **average cross-product of deviations** from the mean.

country	doctor	infmor
Australia	3.8	3.6
Austria	5.3	3.5
Belgium	3.2	3.4
Canada	2.8	4.9
Czech Republic	4.1	3.2
Estonia	3.5	2.4
France	3.4	4.5
Germany	4.4	3.8
Hungary	3.5	3.7
Iceland	3.9	2
Ireland	3.3	3.3
Israel	3.3	3.7
Italy	4	3.1
Korea	2.5	3.2
Latvia	3.3	3.6
Lithuania	4.6	3.7
Mexico	2.4	14.2
New Zealand	3.4	4.7
Norway	5	2.4
Slovak Republic	3.6	5.8
Slovenia	3.3	2.1
Spain	4.4	3.1
Switzerland	4.3	4
Turkey	1.9	10
United Kingdom	3	4.3
United States	2.6	6.5
Average	3.57	4.33

$X - \bar{X}$	$Y - \bar{Y}$
0.23	-0.73
1.73	-0.83
-0.37	-0.93
-0.77	0.57
0.53	-1.13
-0.07	-1.93
-0.17	0.17
0.83	-0.53
-0.07	-0.63
0.33	-2.33
-0.27	-1.03
-0.27	-0.63
0.43	-1.23
-1.07	-1.13
-0.27	-0.73
1.03	-0.63
-1.17	9.87
-0.17	0.37
1.43	-1.93
0.03	1.47
-0.27	-2.23
0.83	-1.23
0.73	-0.33
-1.67	5.67
-0.57	-0.03
-0.97	2.17

$(X - \bar{X})(Y - \bar{Y})$
-0.170
-1.445
0.345
-0.435
-0.602
0.134
-0.028
-0.444
0.044
-0.772
0.279
0.171
-0.532
1.213
0.198
-0.654
-11.535
-0.062
-2.768
0.045
0.602
-1.026
-0.245
-9.457
0.020
-2.099

$$\frac{1}{N-1} \sum (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{-29.22}{26-1} = -1.17$$

Basic Properties of Covariance

- **Symmetry**

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

- **Relationship to variance**

$$\text{Cov}(X, X) = \text{Var}(X)$$

- **Range**

Covariance can take any value from $(-\infty, +\infty)$.

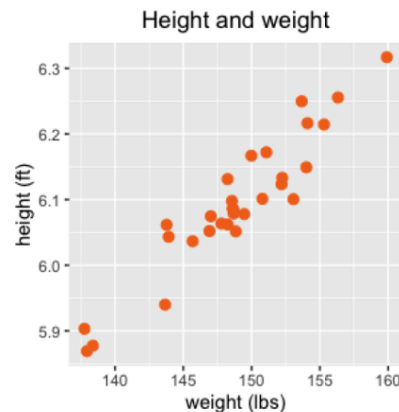
- **Unit dependence**

The magnitude of covariance depends on the units of X and Y , so it is not inherently interpretable without reference to their scales.

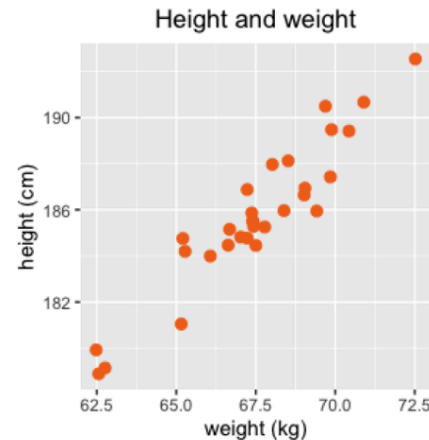
- **Interpretation**

- Positive covariance → variables increase/decrease together
- Negative covariance → variables move in opposite directions
- Zero → no linear relationship

Suppose you are collaborating with a researcher in Europe and you each plot height and weight for the same 30 individuals, but using different units.



Covariance = 7.16



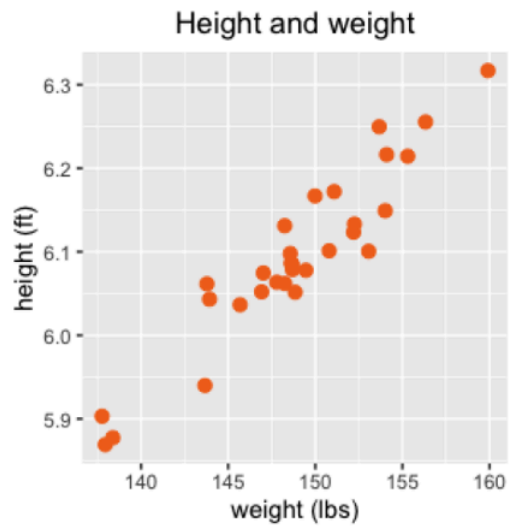
Covariance = 0.52

- The covariance values differ even though the underlying data (the people measured) are identical.
- This occurs because covariance is unit-dependent — it scales with the measurement units of the variables.
- Therefore, covariance by itself is not standardized and not directly comparable across studies or contexts unless the variables are measured in the same units.
- Covariance tells us only the direction of the relationship (positive or negative), not its magnitude on a common scale.

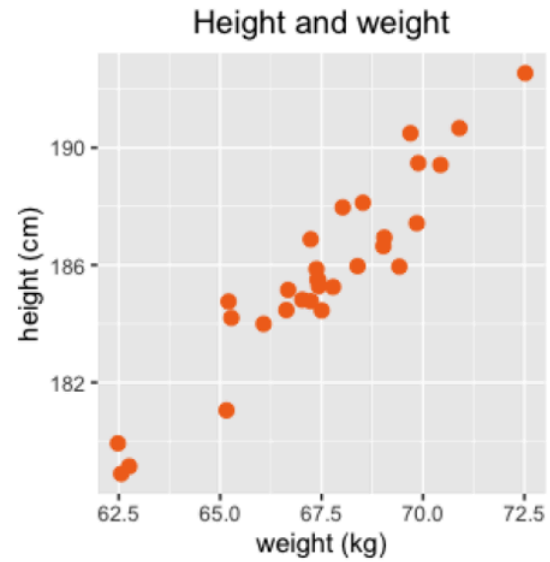
Correlation

$$r = \frac{\frac{1}{N-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})}{\hat{\sigma}_x \hat{\sigma}_y}$$

- Correlation is a standardized form of covariance, obtained by dividing covariance by the product of the standard deviations of the two variables.
- Because it removes the influence of measurement units, correlation is independent of the scale or metric of both variables.
- The correlation coefficient ranges from **-1 to +1**:
 - +1: perfect positive linear association
 - -1: perfect negative linear association
 - 0 : no linear association
- Because correlation is unitless and bounded, it is easier to interpret than covariance.



Correlation = 0.94



Correlation = 0.94

$$\frac{1}{N-1} \sum (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{-29.22}{26-1} = -1.17$$

$$r = \frac{\frac{1}{N-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})}{\hat{\sigma}_x \hat{\sigma}_y} = \frac{-1.17}{0.81 \times 2.57} = -0.563$$

LOCATION	doctor	infmor
Australia	3.8	3.6
Austria	5.3	3.5
Belgium	3.2	3.4
Canada	2.8	4.9
Czech Republic	4.1	3.2
Estonia	3.5	2.4
France	3.4	4.5
Germany	4.4	3.8
Hungary	3.5	3.7
Iceland	3.9	2
Ireland	3.3	3.3
Israel	3.3	3.7
Italy	4	3.1
Korea	2.5	3.2
Latvia	3.3	3.6
Lithuania	4.6	3.7
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New Zealand	3.4	4.7
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Slovak Republic	3.6	5.8
Slovenia	3.3	2.1
Spain	4.4	3.1
Switzerland	4.3	4
Turkey	1.9	10
United Kingdom	3	4.3
United States	2.6	6.5
Average	3.57	4.33
Standard Deviation	0.81	2.57

**GUESS THE
CORRELATION**

**NEW GAME
TWO PLAYERS
SCORE BOARD
ABOUT
SETTINGS**

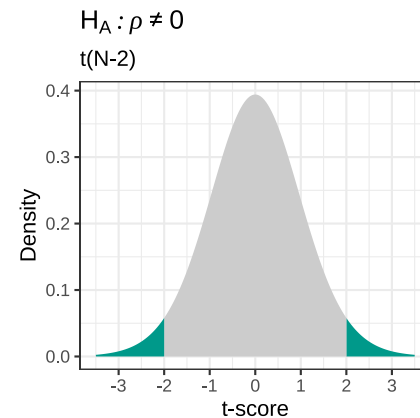
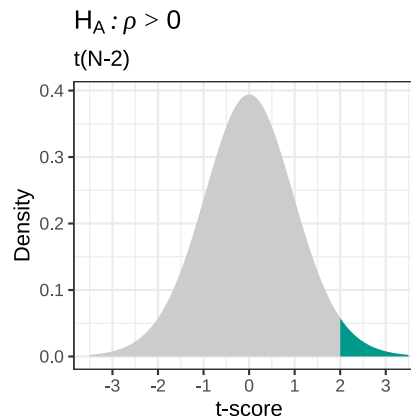
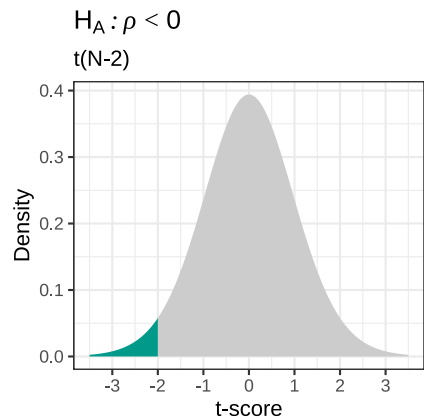
Statistical Inference for Correlation

- We can test whether or not the correlation coefficient is significantly different than zero at the population level.

$$H_0 : \rho = 0$$

- The sampling distribution of sample correlation is a t -distribution with $N-2$ degrees of freedom.

$$t = r \times \sqrt{\frac{N-1}{1-r^2}}$$



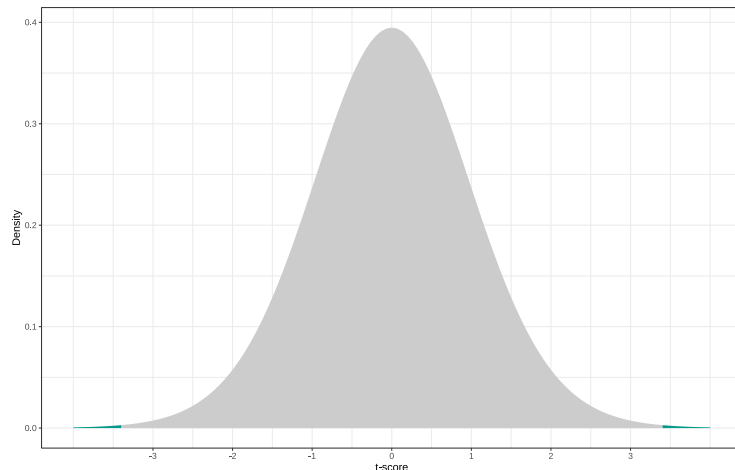
The correlation between the total number of physicians and under-five mortality rate was found to be -0.563 for 26 OECD countries.

Is this significantly different than zero?

$$H_0 : \rho = 0$$

$$H_A : \rho \neq 0$$

$$t = (-0.563) \times \sqrt{\frac{26-1}{1-(-0.563)^2}} = -3.40$$



```
pt(q = 3.4,  
   df=24,  
   lower.tail=FALSE)*2
```

```
#> [1] 0.002357594
```

There was a significant relationship in 2019 between the total number of physicians per 1,000 inhabitant and under-five mortality rate per 1000 live births, $r = -0.56$, $t(24) = -3.4$, $p = 0.0024$.

You can calculate correlation between two variables by using the `cor()` function from the base R. No extra package needed.

But, it only calculates the correlation coefficient and doesn't provide any inferential statistics for its significance.

```
# Import the dataset
```

```
d ← read.csv(file = here('./resources/datasets/infant_mortality.csv'),  
             header = TRUE)
```

```
head(d)
```

	country	doctor	infmor
1	Australia	3.8	3.6
2	Austria	5.3	3.5
3	Belgium	3.2	3.4
4	Canada	2.8	4.9
5	Czech Republic	4.1	3.2
6	Estonia	3.5	2.4

```
cor(x = d$doctor,  
    y = d$infmor)
```

```
[1] -0.5625062
```


Alternatively, you can use the `rcorr()` function from the `Hmisc` package.

```
# Import the dataset
d <- read.csv(file = here('./resources/datasets/infant_mortality.csv'),
              header = TRUE)

head(d)
```

```
  country doctor infmor
1 Australia   3.8   3.6
2 Austria    5.3   3.5
3 Belgium    3.2   3.4
4 Canada     2.8   4.9
5 Czech Republic 4.1   3.2
6 Estonia    3.5   2.4
```

```
# install.packages("Hmisc")

require(Hmisc)

rcorr(x = d$doctor,
      y = d$infmor,
      type="pearson")
```

```
      x      y
x  1.00 -0.56
y -0.56  1.00
```

```
n= 26
```

```
P
      x      y
x  0.0028
y  0.0028
```

It reports the p-value from the *t*-test of significance.

It doesn't report the *t*-statistic or degrees of freedom, so you should compute them.

Correlation Matrix

For a dataset with n numeric variables:

- There exists a $n \times n$ symmetric correlation matrix summarizing all pairwise correlations among the variables.
- The diagonal elements of the correlation matrix are always equal to 1, because the correlation of any variable with itself is perfect.
- The upper and lower off-diagonal elements are identical, since correlation is a symmetric measure, $\rho_{ij} = \rho_{ji}$.
- Therefore, only $\frac{n(n-1)}{2}$ unique correlation estimates are needed to describe all pairwise relationships.
- For example, with five numeric variables, the full 5×5 correlation matrix is:

$$\mathbf{R} = \begin{bmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} & \rho_{15} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} & \rho_{25} \\ \rho_{31} & \rho_{32} & 1 & \rho_{34} & \rho_{35} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 & \rho_{45} \\ \rho_{51} & \rho_{52} & \rho_{53} & \rho_{54} & 1 \end{bmatrix}$$

Example: Life Expectancy dataset

Suppose we select the following five variables from a life expectancy dataset:

- `adult_mortality`: number of deaths between ages 15–60 per 1,000 population
- `under_five_deaths`: number of deaths under age 5 per 1,000 population
- `measles`: reported cases of measles per 1,000 population
- `polio`: polio (Pol3) immunization coverage among 1-year-olds (%)
- `diphtheria`: diphtheria–tetanus–pertussis (DTP3) immunization coverage among 1-year-olds (%)

```
d <- read.csv(file = here('./resources/datasets/life_expectancy.csv'),
              header = TRUE)

d_sub <- d[,c('adult_mortality', 'under_five_deaths', 'measles', 'polio', 'diphtheria')]

my.corr <- cor(x = d_sub)

my.corr
```

```
      adult_mortality under_five_deaths      measles      polio
adult_mortality      1.00000000      0.2101702  0.0550505884 -0.37858149
under_five_deaths      0.21017023      1.00000000  0.7657338626 -0.14648172
measles                0.05505059      0.7657339  1.0000000000 -0.02753727
polio                  -0.37858149     -0.1464817 -0.0275372744  1.00000000
diphtheria             -0.32797139     -0.1370574 -0.0009479256  0.66094649
      diphtheria
adult_mortality -0.3279713856
under_five_deaths -0.1370574213
measles          -0.0009479256
polio            0.6609464929
diphtheria       1.0000000000
```

```
round(my.corr,2)
```

```
      adult_mortality under_five_deaths measles polio diphtheria
adult_mortality      1.00      0.21      0.06 -0.38      -0.33
under_five_deaths      0.21      1.00      0.77 -0.15      -0.14
measles                0.06      0.77      1.00 -0.03      0.00
polio                  -0.38     -0.15     -0.03  1.00      0.66
diphtheria             -0.33     -0.14     0.00  0.66      1.00
```

When you need to test the significance of the correlations, you can use the `rcorr()` function from the `Hmisc` package.

```
my.corr <- rcorr(x = as.matrix(d_sub))

# 5 x 5 correlation matrix

my.corr$r

      adult_mortality under_five_deaths      measles      polio
adult_mortality      1.00000000      0.2101702      0.0550505884 -0.37858149
under_five_deaths      0.21017023      1.00000000      0.7657338626 -0.14648172
measles                0.05505059      0.7657339      1.000000000000 -0.02753727
polio                  -0.37858149     -0.1464817     -0.0275372744      1.00000000
diphtheria             -0.32797139     -0.1370574     -0.0009479256      0.66094649
      diphtheria
adult_mortality -0.3279713856
under_five_deaths -0.1370574213
measles          -0.0009479256
polio            0.6609464929
diphtheria       1.0000000000

# Sample size for each correlation

my.corr$n

      adult_mortality under_five_deaths measles polio diphtheria
adult_mortality      180              180      180      180      180
under_five_deaths      180              180      180      180      180
measles                180              180      180      180      180
polio                  180              180      180      180      180
diphtheria             180              180      180      180      180

# p-values for significance of each correlation

my.corr$p

      adult_mortality under_five_deaths      measles      polio
adult_mortality      NA              0.004628915      0.4629533      1.604153e-07
under_five_deaths      4.628915e-03              NA      0.0000000      4.974338e-02
measles                4.629533e-01              0.0000000000      NA      7.136585e-01
polio                  1.604153e-07              0.049743378      0.7136585      NA
diphtheria             6.973657e-06              0.066553801      0.9899237      0.000000e+00
      diphtheria
adult_mortality      6.973657e-06
under_five_deaths      6.655380e-02
measles                9.899237e-01
polio                  0.000000e+00
diphtheria            NA
```

APA Format Reporting of Correlations

Sample Correlation Table

Table 1

Descriptive Statistics and Correlations for Study Variables

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	1	2	3	4	5
1. Adult Mortality ^a	180	151.36	96.60	—				
2. Under Five Deaths ^b	180	31.60	108.59	.21**	—			
3. Measles ^c	180	1527.28	7905.53	.06	0.77***	—		
4. Polio ^d	180	83.10	24.41	-.38***	-.15*	-.03	—	
5. Diphtheria ^e	180	84.56	22.33	-.33***	-.14	.00	.66***	—

^a Number of deaths between ages 15–60 per 1,000 population in 2015

^b Number of deaths under age 5 per 1,000 population in 2015

^c reported cases of measles per 1,000 population in 2015

^d polio (Pol3) immunization coverage among 1-year-olds (%) in 2015

^e diphtheria–tetanus–pertussis (DTP3) immunization coverage among 1-year-olds (%) in 2015

* $p < .05$. ** $p < .01$. *** $p < .001$.

Handling missing data when computing correlations

Observed Sample

V1	V2	V3
5	NA	9
3	2	NA
4	8	5
NA	3	NA
1	2	6
NA	9	4
NA	5	NA
3	NA	8
2	6	5
NA	8	4

There are two ways dealing with missing data when computing correlations

- **Listwise deletion**
- **Pairwise deletion**

Listwise deletion: any row with a missing value is removed from data and then the correlations are computed.

Observed Sample

V1	V2	V3
5	NA	9
3	2	NA
4	8	5
NA	3	NA
1	2	6
NA	9	4
NA	5	NA
3	NA	8
2	6	5
NA	8	4

Analytical Sample

V1	V2	V3
4	8	5
1	2	6
2	6	5

Pairwise deletion: rows with missing data are removed per pair of variables, then the correlation is computed for each pair.

Observed Sample

V1	V2	V3
5	NA	9
3	2	NA
4	8	5
NA	3	NA
1	2	6
NA	9	4
NA	5	NA
3	NA	8
2	6	5
NA	8	4

Analytical Sample
for V1 and V2

V1	V2
3	2
4	8
1	2
2	6

Analytical Sample
for V1 and V3

V1	V3
5	9
4	5
1	6
3	8
2	5

Analytical Sample
for V2 and V3

V2	V3
8	5
2	6
9	4
6	5
8	4

Handling Missing Data in Correlation Analysis

Listwise Deletion

- **Pro:**
All correlations are calculated using the **same complete set of cases**, ensuring consistency across variable pairs.
- **Con:**
May substantially **reduce sample size**, especially when multiple variables contain missing values.
This loss of data can decrease statistical power.

Pairwise Deletion

- **Pro:**
Uses all available data for each variable pair, preserving as many observations as possible for correlation calculations, thereby minimizing data loss
- **Con:**
Different correlations may rely on **different subsets of observations**, leading to inconsistency across the correlation matrix.
Results may also be **biased** if the missingness is not random (not MCAR).

Example: Life Expectancy dataset

Suppose we select the following five variables from a life expectancy dataset:

- `life_expectancy`: life expectancy in age
- `bmi`: average Body Mass Index of entire population
- `percentage_expenditure`: Expenditure on health as a percentage of Gross Domestic Product per capita(%)
- `gdp`: Gross Domestic Product per capita (in USD)
- `hepatitis_b`: Hepatitis B (HepB) immunization coverage among 1-year-olds (%)

Working with `cor()` function

```
d <- read.csv(file = here('./resources/datasets/life_expectancy.csv'),
              header = TRUE)

d_sub <- d[,c('life_expectancy', 'bmi', 'percentage_expenditure', 'gdp', 'hepatitis_b')]

my.corr <- cor(x = d_sub)

round(my.corr, 2)
```

```
      life_expectancy bmi percentage_expenditure gdp
life_expectancy      1.00 NA                  0.04 NA
bmi                  NA  1                   NA    NA
percentage_expenditure 0.04 NA                  1.00 NA
gdp                  NA NA                   NA     1
hepatitis_b          NA NA                   NA    NA

      hepatitis_b
life_expectancy NA
bmi              NA
percentage_expenditure NA
gdp              NA
hepatitis_b      1
```

If your dataset contains missing values, you must specify how the `cor()` function should handle them when computing correlations.

Listwise deletion

```
my.corr <- cor(x = d_sub,  
              use = 'complete' )  
  
round(my.corr,2)
```

```
      life_expectancy  bmi percentage_expenditure  gdp  
life_expectancy      1.00 0.56                    0.06 0.47  
bmi                  0.56 1.00                    0.05 0.37  
percentage_expenditure 0.06 0.05                    1.00 -0.03  
gdp                  0.47 0.37                   -0.03 1.00  
hepatitis_b          0.41 0.19                    0.05 0.14  
  
      hepatitis_b  
life_expectancy  0.41  
bmi              0.19  
percentage_expenditure 0.05  
gdp               0.14  
hepatitis_b      1.00
```

However, `cor()` does not report the number of observations retained after listwise deletion.

To see how many observations were actually used:

```
nrow(na.omit(d_sub))
```

```
[1] 144
```

This command returns the number of complete cases in your dataset.

Pairwise deletion

```
my.corr <- cor(x = d_sub,  
              use = 'pairwise' )  
  
round(my.corr,2)
```

```
life_expectancy    life_expectancy  bmi  percentage_expenditure  gdp  
life_expectancy    1.00 0.51  
bmi                0.51 1.00  
percentage_expenditure 0.04 0.04  
gdp                0.45 0.35  
hepatitis_b        0.43 0.18  
                    hepatitis_b  
life_expectancy    0.43  
bmi                0.18  
percentage_expenditure 0.04  
gdp                0.14  
hepatitis_b        1.00
```

`cor()` does not report the number of observations used for each pair of variables.

You can write some custom code to get that information, but it requires some programming.

Working with `rcorr()` function

`rcorr()` function always use pairwise deletion in the presence of missing data.

```
my.corr <- rcorr(x = as.matrix(d_sub))
```

```
my.corr
```

```
life_expectancy    life_expectancy    bmi    percentage_expenditure    gdp
life_expectancy    1.00 0.51
bmi                 0.51 1.00
percentage_expenditure 0.04 0.04
gdp                 0.45 0.35
hepatitis_b         0.43 0.18
                    0.04 0.14
```

```
life_expectancy    hepatitis_b
bmi                 0.43
percentage_expenditure 0.18
gdp                 0.04
hepatitis_b         0.14
                    1.00
```

n

```
life_expectancy    life_expectancy    bmi    percentage_expenditure    gdp
life_expectancy    180 178
bmi                 178 178
percentage_expenditure 180 178
gdp                 154 152
hepatitis_b         171 169
                    171 146
```

```
life_expectancy    hepatitis_b
bmi                 171
percentage_expenditure 169
gdp                 171
hepatitis_b         146
                    171
```

P

```
life_expectancy    life_expectancy    bmi    percentage_expenditure    gdp
life_expectancy    0.0000    0.0000    0.5491    0.0000
bmi                 0.0000    0.6256    0.0000    0.0000
percentage_expenditure 0.5491    0.6256    0.7045    0.7045
gdp                 0.0000    0.0000    0.7045    0.0871
hepatitis_b         0.0000    0.0167    0.5890    0.0871
```

```
life_expectancy    hepatitis_b
bmi                 0.0000
percentage_expenditure 0.0167
gdp                 0.5890
hepatitis_b         0.0871
```